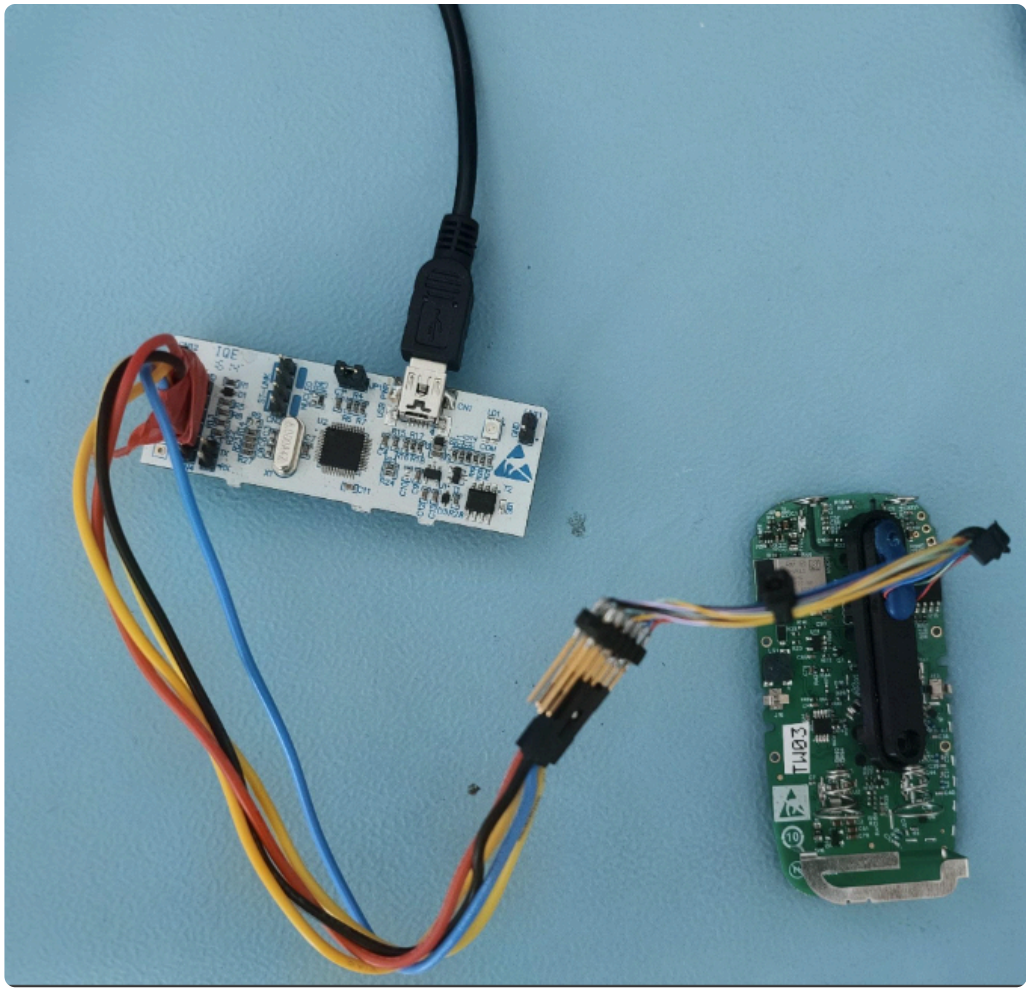
Sidewalk DUT setup for radiated emission testing

Firmware update

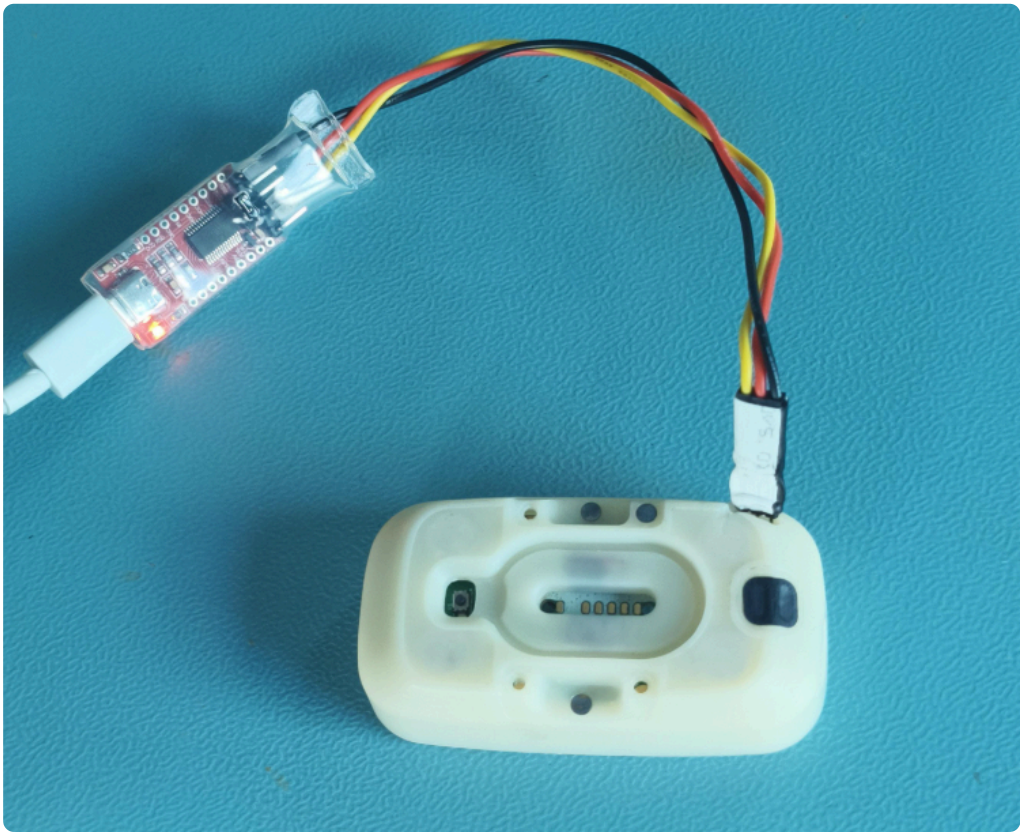
There are two MCUs that needs firmware - **main MCU** STM32WLE5(SubGHz radio) and **Bluetooth MCU** STM32WBA5. Both can be programed by the client application or by HVF test application from TxWireless. As there is only one UART port easily accessible on the board edge and it is the main MCU UART the access to the Bluetooth MCU UART was achieved by entering main MCU UAURT into passthrough mode using HVF command. At the time of shipment of the two devices for certification the main MCU HVF firmware was not handling properly all of the commands required for Bluetooth radiated emissions testing. Before going to the RF test house the main MCU firmware and Bluetooth MCU must be updated to the latest version. To do that use the proprietary pogopin connector attached to the ST_Link programmer. If needed contact TxWireless for more details.



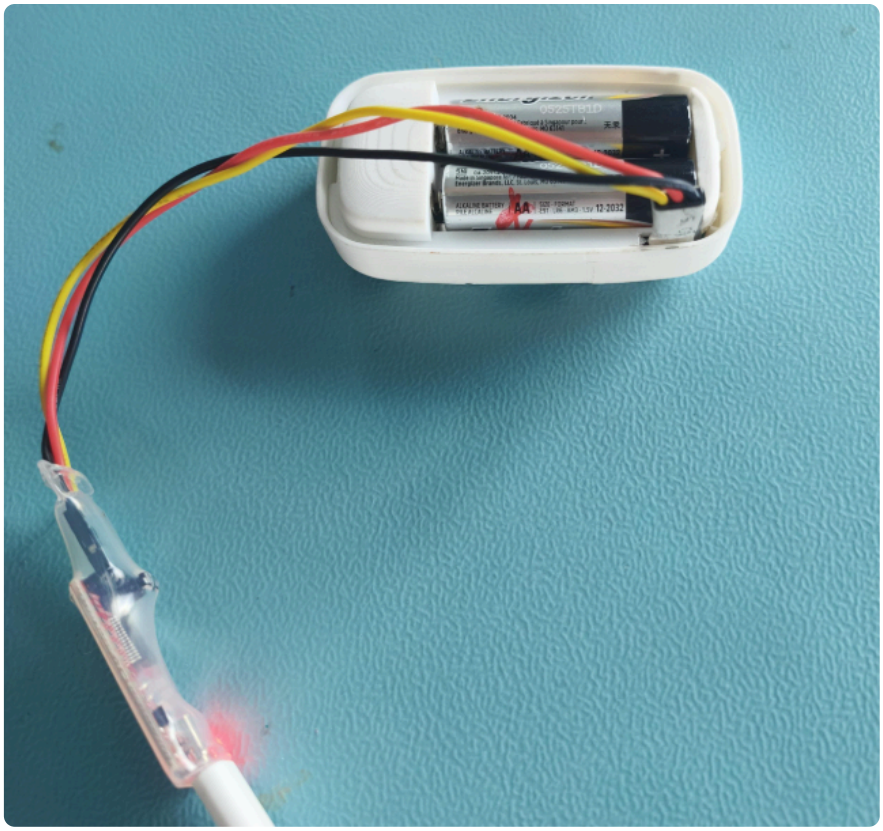
STM32WLE5 main MCU firmware updated

DUT radio configuration

For the emissions testing the device must be enclosed in the final enclosure running from batteries, the radio must be configured for RF transmission with specified power, frequency, modulation and bandwidth. To do that TxWireless made a small hole(2*6mm) in the enclosure to access the main MCU chip STM32WLE5 3-pin UART communication port on the PCB. There are two ways to do that.



UART access hole in the bottom part of the enclosure



UART access hole in the mid frame part of the enclosure

With such UART(115200,8n1) access to the main MCU running HVF firmware we can setup RF transmission parameters. Use any serial terminal console software running on a PC or 'Serial USB Terminal' app on Android smartphone.

- for SubGHz radio directly through the HVF commands:
  - **rf_pwr_set 20** //output power in dBm
  - **rf_freq_set 915000000** //Carrier frequency
  - modulation
    - LoRa
      - **rf_modulation_set 1** //LoRa
      - **rf_bandwith_set 5** //LoRa Bandwidth select 0=7.8125 kHz, 1=15.625 kHz, 2=31.25 kHz, 3=62.5 kHz, 4=125 kHz, 5=250 kHz, 6=500 kHz
    - FSK
      - **rf_modulation_set 0** //FSK
      - **rf_bandwidth_set 50000** //FSK bandwidth select from 4800 to 467000bps
  - **rf_tx_test 50000 10** //50000 RF bursts every 10ms - overall test length is about 20 hours for LoRa and 7 hours for FSK

```
13:59:02.778 rf_pwr_set 20
13:59:02.788 rf_pwr_set 20
13:59:02.788 TX power set to 20 dBm
13:59:02.788 <[OK]>^M
13:59:02.788 hvf> 13:59:03.781 rf_freq_set 915000000
13:59:03.792 rf_freq_set 915000000
13:59:03.792 Operating frequency set to 915000000 Hz
13:59:03.793 <[OK]>^M
13:59:03.793 hvf> 13:59:04.786 rf_modulation_set 1
13:59:04.796 rf_modulation_set 1
13:59:04.796 LoRa modulation set
13:59:04.796 <[OK]>^M
13:59:04.796 hvf> 13:59:05.791 rf_bandwidth_set 2
13:59:05.801 rf_bandwidth_set 2
13:59:05.801 LoRa bandwidth set to 31.25 kHz
13:59:05.801 <[OK]>^M
13:59:05.801 hvf> 13:59:06.795 rf_tx_test 50000 10
13:59:06.802 rf_tx_test 50000 10
13:59:06.803 Tx Test
13:59:06.803
13:59:06.822 Tx 1 of 50000
13:59:06.822
13:59:13.419 OnTxDone
13:59:13.419
13:59:13.430 Tx 2 of 50000
```

- for Bluetooth radio indirectly through the main MCU passthrough UART:
  - **ipc_passthrough** //enters main MCU UART bridge to BLE chip
  - **ble_cwtx_start** *channel bandwidth power freq_offset //setup Bluetooth radio*
    - channel 0-39 //
    - bandwidth 1m //1m -1MHz, 2m - 2MHz
    - power in dBm// 10dBm is maximum for Bluetooth chip STM32WBA5
    - frequency offset 0



-

After configuring DUT radio chip into RF transmission the UART cable can be detached, device enclosed and placed into the RF chamber.

To stop RF transmission or reconfigure radio please remove one of the batterie for few seconds - this will power cycle the device.

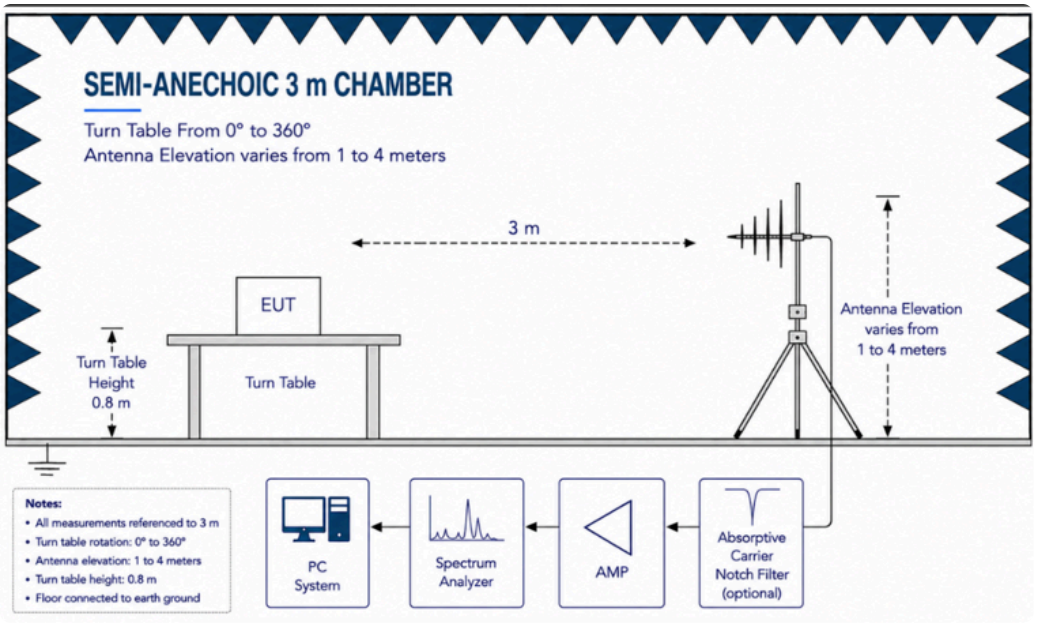
Perform radiated emissions tests separately for the SubGHz radio and for the Bluetooth radio.

One set of batteries should last for about 4 hours.

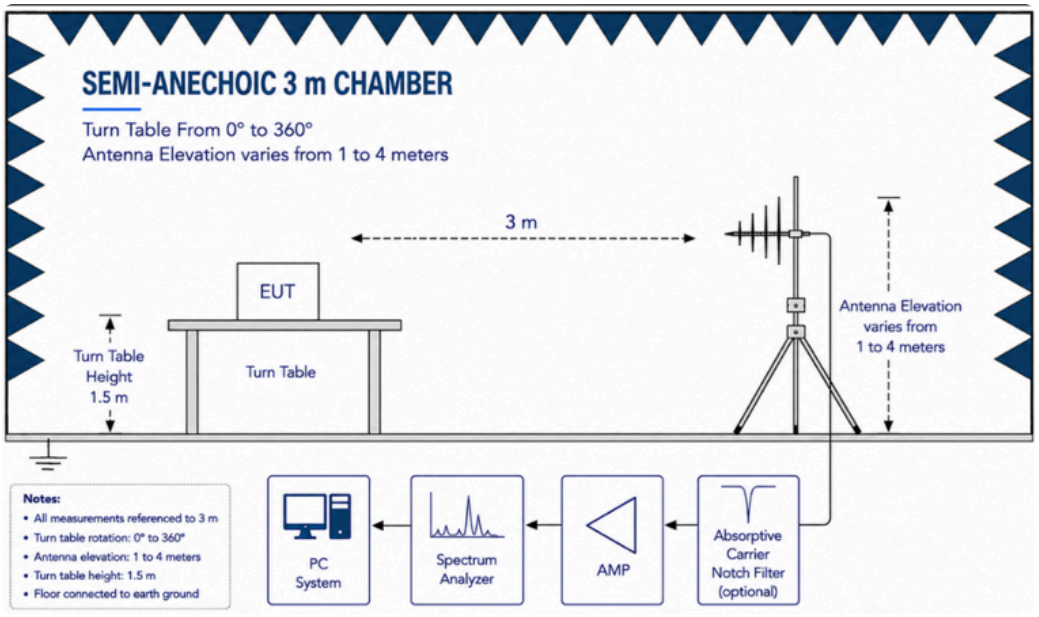
RF chamber setup

During measurements please follow American National Standard of Procedures for Compliance Testing of Unlicensed Wireless Devices ANSI C63.10-2020（IEEE C63.10-2020).

Testing below 1GHz



Testing above 1GHz



Device under test is transmitting +20dBm output power at carrier frequency 915MHz. Please ensure the measurement receiver is not saturated by such a signal when measuring emissions above 1GHz, due to close proximity of the strong carrier signal(0.915GHz/+20dBm) to the measurement range. Consider increasing attenuation or use absorptive high-pass filter or absorptive Notch filter before pre-amp to limit the carrier power if necessary. For more information about the issue and how to check it please refer to the chapter nr 5 of the guidelines below www.tekbox.com/product/AN_spectrum_analyzers_for_EMC_testing.pdf